7. Let $x(t)$ be a sampled signal specified as

$$x(t) = \sum_{n=0}^{\infty} e^{-nT}\delta(t - nT)$$

$x(t) = \delta(t) + e^{-T}\delta(t-T) + e^{-2T}\delta(t-2T) + \cdots$

where T > 0

(a) Determine X(s) and the ROC

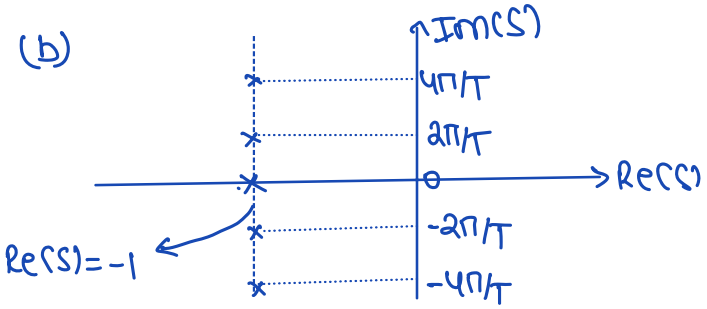
(b) Sketch the pole zero plot for X(s)

(a) $X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt$

$= 1 + e^{-(s+1)T} + e^{-(s+1)2T} + \cdots = \frac{1}{1 - e^{-(s+1)T}}$

$|s| > 1$

(b)



$e^{(s+1)T} = 1$

$(s+1)T = 2\pi j k$

$s = -1 + \frac{2\pi j k}{T}$

$Im(s) = 2\pi k/T$